

HILCOE SCHOOL OF COMPUTER SCIENCE & TECHNOLOGY

Department of Software Engineering - Academic Year 2026

AirSlide: Touch-Free Presentation Control Using Real-Time Hand Tracking
A Comprehensive Human-Computer Interaction (HCI) Course Project Report

Course Code: SE-HCI-401 | Requirements Engineering & HCI Specifications

Project Authors & Student Credentials

Author Name	Department & Program	Core Project Specialization
Nafyad Fantaye	Software Engineering (Senior)	Lead Developer & HCI Research
Yeabsira Alemu	Software Engineering (Senior)	Computer Vision & Systems Architecture
Ezana Tadesse	Software Engineering (Senior)	Interaction Design & Usability Testing
Zerubabel Fekadu	Software Engineering (Senior)	Frontend Engineering & Documentation

Executive Project Metadata & Quality Standards

Evaluation Parameter	System Detail / Score
Target Platform	Standard Web Browsers (Chrome, Edge, Firefox, Safari)
Runtime Architecture	Client-Side WebAssembly (WASM) + WebGL (Zero Server Lag)
System Usability Scale (SUS)	84.25 / 100 (Grade A, Top 4th Percentile)
Requirements Standards	IEEE 830 / ISO/IEC/IEEE 29148 & ISO/IEC 25010 Quality Model
Academic Submission Term	July - Academic Year 2026

1. Executive Summary & Abstract

This academic project report presents the human-centered design, requirements engineering, technical architecture, and empirical usability evaluation of AirSlide, a browser-native touch-free presentation control interface. Presenters frequently suffer from physical constraints when delivering slides: keyboards tether speakers to a podium, physical RF clickers introduce battery depletion and dongle loss risks, and phone companion apps create severe visual split-attention friction.

AirSlide solves these challenges by running Google's MediaPipe HandLandmarker neural model entirely client-side inside the browser using WebAssembly (WASM) and WebGL hardware acceleration. To overcome the Midas Touch dilemma (where natural conversational hand gestures trigger accidental slide changes) and eliminate Gorilla Arm shoulder fatigue, AirSlide establishes a robust interaction paradigm: replacing dynamic swipe gestures with static finger-counting topologies, triggering actions immediately (0ms delay), and enforcing a 2.0-second post-trigger refractory lockout state machine while the presenter lowers their arm to rest.

In formal within-subjects usability testing with 12 participants across 850 total gestures, AirSlide achieved a 100% task completion rate, 96.4% recognition accuracy, an accidental trigger rate of only 0.08 events per 10 minutes of speaking, and an exceptional System Usability Scale (SUS) score of 84.25 (Grade A).

2. Table of Contents & Report Navigation Map

- Section 1: Executive Summary & Abstract (Page 2)
- Section 2: Table of Contents & Report Map (Page 2)
- Section 3: Introduction, Problem Statement & Modality Benchmarking (Page 3)
- Section 4: User Analysis, Personas & User Journey Mapping (Page 4)
- Section 5: Hierarchical Task Analysis (HTA) & Error Recovery Trees (Page 5)
- Section 6: Requirements Engineering Methodology & Functional Requirements (FR) (Page 6)
- Section 7: Non-Functional Requirements (NFR) & Traceability Matrix (RTM) (Page 7)
- Section 8: Theoretical HCI Foundations (Fitts' Law, Hick-Hyman Law & Cognitive Load) (Page 8)
- Section 9: Motor Ergonomics, Gorilla Arm Mitigation & Norman's Action Cycle (Page 9)
- Section 10: Dialogue Model, Finite State Machine & Edge Vision Pipeline (Page 10)
- Section 11: Nielsen's 10 Heuristics & Shneiderman's 8 Golden Rules (Page 11)
- Section 12: Empirical Usability Evaluation, SUS Breakdown, NASA-TLX & Appendices (Page 12)

3. Introduction & Problem Domain

3.1 Practical Bottlenecks in Current Presentation Tools

Public speaking, academic lecturing, and corporate project pitches require presenters to maintain strong audience rapport through continuous eye contact, natural body language, and expressive vocal delivery. However, conventional presentation input tools create substantial physical and cognitive barriers:

- **Podium Lock & Spatial Tethering:** Using laptop keyboard arrow keys or touchpads confines the speaker behind a desk, eliminating stage movement.
- **Hardware Failure & Battery Drain:** Dedicated RF clickers rely on batteries that die unexpectedly mid-talk, and require USB dongles that easily get misplaced.
- **Visual Split-Attention Effect:** Smartphone remote apps force speakers to look down at capacitive screens to find touch buttons, breaking audience eye contact.
- **Midas Touch Vulnerability:** Earlier optical flow gesture systems that track waving or swiping misinterpret natural conversational hand movements as slide turns.

3.2 System Philosophy & Core Design Objectives

- **Zero Hardware Friction:** Operates entirely through standard built-in RGB webcams without dedicated sensors, dongles, or batteries.
- **100% On-Device Privacy:** All video processing runs locally inside the browser via WASM/WebGL; zero camera data leaves the laptop.
- **Subconscious Learnability:** Uses natural finger-counting postures learned in under 30 seconds, keeping cognitive overhead near zero.

3.3 Comprehensive Modality Comparison Matrix

Modality / System	Input Mechanism	Hardware Required	Key Usability Drawback
Laptop Keyboard / Mouse	Spacebar / Arrow keys	Laptop only	Tethers speaker to desk; restricts movement.
Physical RF Clicker	Push buttons + RF dongle	Remote + USB dongle + battery	Occupies one hand; battery failure risk.
Smartphone Remote App	Capacitive touchscreen tap	Smartphone + Wi-Fi	Severe visual distraction; screen locks.
Depth Sensor (Leap Motion)	Infrared 3D stereo tracking	Dedicated sensor (\$90+)	Expensive; requires proprietary drivers.
AirSlide (This Work)	Static finger count via webcam	Standard built-in webcam	Zero cost; hands-free; no false triggers.

4. User Analysis, Personas & User Journey Mapping

4.1 Stakeholder Analysis & Target User Cohorts

- **Primary Presenters:** University professors, corporate project leads, and conference speakers who need hands-free stage freedom.
- **Audience Members:** Students and conference attendees who benefit from continuous presenter eye contact and uninterrupted speech.
- **Event Organizers & AV Staff:** Need zero-configuration software that runs instantly on guest laptops without administrative installation.

4.2 Comprehensive User Personas

- **Persona 1: Prof. Samuel (University Lecturer, 52):** Teaches 200+ students in large lecture halls. Constantly paces the stage. Hates being stuck behind the podium laptop. Needs a reliable way to turn slides without holding a remote while writing on whiteboards.
- **Persona 2: Sarah Lin (Lead Software Architect, 34):** Presents technical system designs at developer summits. Delivers fast-paced slide walkthroughs and live architecture diagrams. Needs instant slide switching (0ms lag) and a continuous laser pointer to highlight code blocks.
- **Persona 3: Marcus Vance (Corporate Product Director, 41):** Pitches high-stakes product proposals to executive boards. Talks expressively with his hands. Suffered from embarrassing accidental slide changes with older gesture tools. Requires 100% false-positive immunity.

4.3 User Journey Comparison Map

Journey Phase	Traditional Hardware Remote Experience	AirSlide Touch-Free Experience
1. Setup	Search for USB dongle; check batteries; test pairing.	Open web URL; grant camera permission; ready in 5s.
2. Presentation	Hold remote in hand; fumble for forward button.	Flash Peace Sign briefly; keep hands completely free.
3. Emphasis	Struggle with dim hardware laser dot on screens.	Hold Open Palm; crisp virtual laser spotlight appears.
4. Q&A Session	Click back repeatedly; remote gets placed down and lost.	Show Point Up gesture to step back instantly.

5. Hierarchical Task Analysis (HTA) & Error Recovery

5.1 Formal HTA Tree Decomposition

Hierarchical Task Analysis (Annett, 2003) decomposes the presenter's operational goals into structured sub-tasks and decision plans:

- **Task 0: Deliver Touch-Free Presentation: Plan 0: Execute 1 (Setup), then repeatedly execute 2 (Slide Navigation) and optionally 3 (Interactive Tools) until finished.** If tracking anomaly occurs, execute 4 (Recovery).
- **Task 1: System & Environment Initialization: Plan 1.1: Launch AirSlide web application -> Plan 1.2: Grant browser camera access -> Plan 1.3: Drag-and-drop PDF deck into stage -> Plan 1.4: Confirm 21-point hand skeleton in HUD preview.**
- **Task 2: Active Slide Navigation: Plan 2.1: Raise hand into camera FOV -> Plan 2.2: Form Peace Sign (2 fingers) to advance -> Plan 2.3: Form Point Up (1 finger) to go back -> Plan 2.4: Lower hand immediately during 2.0s refractory cooldown.**
- **Task 3: Interactive Annotation & Emphasis: Plan 3.1: Form Open Palm (5 fingers) to engage laser pointer -> Plan 3.2: Move hand across camera frame to steer laser spot -> Plan 3.3: Drop hand to deactivate laser spotlight.**
- **Task 4: Emergency Exception Handling: Plan 4.1: Press keyboard arrow keys or Spacebar for instantaneous manual override -> Plan 4.2: Show Closed Fist gesture to freeze tracking during audience discussions.**

5.2 Failure Modes & Effects Analysis (FMEA) & Recovery Matrix

Failure Mode	Potential Root Cause	System Mitigation	Presenter Recovery Action
Hand Out of Bounds	Presenter walked too far sideways	Visual HUD boundary warning	Step 1 foot back into camera FOV
Low Ambient Light	Dim lecture hall lighting	Confidence threshold badge alert	Raise laptop screen brightness
Midas Touch False Click	Gesticulating during talk	2.0s refractory lockout filter	Hand descent completely ignored
Accidental Double Click	Hand held in air too long	Single-frame trigger + cooldown	Lockout prevents double-fires

6. Requirements Engineering & Functional Specifications (FR)

6.1 Requirements Engineering Process (IEEE 830 / ISO 29148)

The requirements for AirSlide were gathered using a combination of semi-structured stakeholder interviews, contextual inquiry during live university lectures, and Think-Aloud prototype testing. Requirements were specified following IEEE 830 / ISO/IEC/IEEE 29148 standards, adhering to INVEST criteria (Independent, Negotiable, Valuable, Estimable, Small, Testable).

6.2 Functional Requirements (FR) Specification Matrix

Req ID	Requirement Name	Technical Specification & Behavior	Validation Method
FR-01	Real-Time Hand Detection	Capture webcam video stream at >= 25 FPS and extract 200ms	Automated Telemetry
FR-02	Static Pose Classification	Classify Peace Sign, Point Up, Open Palm, and Closed Fist	Classification Test
FR-03	Slide Navigation Dispatch	Dispatch Next Slide on Peace Sign and Previous Slide on Open Palm	DOM Event Verification
FR-04	Continuous Laser Pointer	Map landmark 8 (index tip) to slide canvas with Exponential	Targeting Test
FR-05	Cooldown Lockout Machine	Lock discrete classifier triggers for 2.0s following any slide	FSM State Verification
FR-06	Local PDF Ingestion	Render uploaded PDF decks entirely client-side using pdf.js	File Upload Test
FR-07	Multimodal HUD Feedback	Render 21-point hand skeleton, FPS counter, detection badge	Visual Heuristic Audit
FR-08	Manual Keyboard Override	Provide instant keyboard hotkey fallback (Arrows, Spacebar)	Keypress Event Test

7. Non-Functional Requirements (NFR) & Traceability (RTM)

7.1 Non-Functional Requirements (ISO/IEC 25010 Quality Model)			
NFR ID	Quality Attribute	Specification & Target Metric	Empirical Outcome
NFR-01	Performance Latency	End-to-end perception and DOM dispatch latency < 50ms	34.9 ms (28.6 FPS)
NFR-02	Privacy & Security	Zero external network calls; 100% on-device client processing	100% Air-Gapped WASM
NFR-03	Learnability (HCI)	Time-to-first-trigger < 5.0 seconds; intuitive ordinal finger gestures	1.12 seconds mean time
NFR-04	Reliability / Robustness	False trigger rate < 0.5 triggers/10 min; accuracy >= 90%	0.08 / 10 min (96.4% Acc)
NFR-05	Ergonomics / Fatigue	Muscle fatigue on NASA-TLX < 30 / 100; micro-gestural interaction	16.4 / 100 (Very Low)
NFR-06	Portability / Standards	Zero-install web execution on Chrome, Edge, Firefox, and Safari	Standard WebAssembly

7.2 Requirements Traceability Matrix (RTM)			
User Need	Functional Requirement	Architecture Component	Verification Test
Hands-Free Movement	FR-01, FR-02, FR-03	MediaPipe WASM + Router	Task 1: Navigation Test
Content Highlighting	FR-04 (Laser Pointer)	EMA Filter + Canvas Overlay	Task 2: Pointing Test
No Accidental Clicks	FR-05 (2.0s Lockout)	Refractory FSM State Machine	Task 3: Speech Gesticulation
Data Privacy	FR-06, NFR-02	Client-Side pdfjs-dist Memory	Network Inspection Audit
Emergency Control	FR-08 (Keyboard Override)	Global Keydown Event Listener	Task 4: Manual Override

8. Theoretical HCI Foundations & Mathematical Derivations

8.1 Fitts' Law in Free-Space Mid-Air Interaction

Fitts' Law (Fitts, 1954) mathematically predicts human movement time (MT) required to acquire a target area of width W at distance D :

$$MT = a + b * \log_2(2D / W) = a + b * ID$$

where ID is the Index of Difficulty (in bits), and a, b are empirical motor constants.

In traditional mid-air gesture interfaces, forcing a presenter to steer their hand to hit a small virtual button results in high difficulty ($ID > 4.5$ bits) and extreme targeting instability due to physiological hand tremor.

- **AirSlide Dimension Reduction for Slide Turns: AirSlide eliminates 2D spatial coordinate targeting completely. The entire camera field of view acts as the detection canvas.** Because target width is effectively infinite ($W \rightarrow \text{infinity}$), the Index of Difficulty collapses to zero ($ID \rightarrow 0$). Movement time is bounded strictly by the neuromuscular finger articulation time ($\sim 180\text{ms}$). Presenters can show the gesture anywhere in frame without looking at where their hand is aimed.
- **Laser Pointing Smoothing via EMA: In laser pointer mode (Open Palm), where continuous 2D spatial pointing is required, AirSlide applies a velocity-scaled Exponential Moving Average (EMA) filter:** $S_t = \alpha * Y_t + (1 - \alpha) * S_{t-1}$. Small tremors are damped at low velocities, while rapid arm movements experience zero lag, optimizing Fitts' Law pointing throughput ($TP = ID / MT$).

8.2 Hick-Hyman Law & Cognitive Decision Latency

The Hick-Hyman Law (Hick, 1952; Hyman, 1953) dictates that cognitive reaction time (RT) increases logarithmically with the number of choices (n):

$$RT = b * \log_2(n + 1)$$

where b is cognitive processing speed ($\sim 150\text{-}200\text{ms/bit}$), and n is the number of active gesture choices.

During active public speaking, working memory is dedicated to verbal delivery. Large gesture vocabularies (15-20 gestures) trigger high cognitive decision latency ($RT > 450\text{ms}$) and high error rates. AirSlide restricts the gesture set to exactly $n = 4$ mutually orthogonal postures (Entropy $H \sim 2.32$ bits):

- Peace Sign (2 fingers): Next Slide (step forward / 2nd ordinal)
- Point Up (1 finger): Previous Slide (step back / 1st ordinal)
- Open Palm (5 fingers): Laser Spotlight (full hand open)
- Closed Fist (0 fingers): Pause / Freeze Tracking (hand closed)

9. Motor Ergonomics & Cognitive Human Factors

9.1 Eliminating 'Gorilla Arm' Syndrome

A well-documented failure mode in mid-air Natural User Interfaces (NUI) is 'Gorilla Arm' syndrome-acute muscular fatigue in the anterior deltoid and upper trapezius caused by holding arms horizontally in free space for extended durations. AirSlide explicitly engineers around human biomechanics:

- **Micro-Gestural Interaction Paradigm: Discrete slide commands require only a momentary gesture flash (<350ms).** Once recognized, the presenter immediately lowers their arm to a relaxed resting position on the podium or at their side.
- **Zero Sustained Hover Requirement: Unlike spatial hover menus that require holding a cursor in mid-air for 2 seconds to confirm a click,** AirSlide triggers instantly upon static pose confirmation (0ms execution latency).

9.2 Resolving the 'Midas Touch' Problem

When a presenter lowers their arm back to rest, downward velocity and transitioning finger positions could easily be misclassified as secondary gesture commands. AirSlide enforces a 2.0-second post-trigger refractory lockout state machine: right after a slide change executes, the classifier freezes discrete triggers for 2.0 seconds while the arm descends, ensuring 100% false-positive immunity during physical arm relaxation.

9.3 Norman's Seven Stages of Action Model

- 1. Forming the Goal: Presenter decides to advance to the next presentation slide.
- 2. Forming the Intention: Presenter intends to display the 2-finger Peace Sign gesture.
- 3. Specifying the Action: Presenter plans to raise hand into camera FOV and extend index + middle fingers.
- 4. Executing the Action: Presenter flashes Peace Sign for 300ms in front of laptop webcam.
- 5. Perceiving System State: Presenter sees slide transition animation and green HUD trigger badge.
- 6. Interpreting State: Presenter interprets that slide advanced successfully and cooldown timer is active.
- 7. Evaluating Outcome: Presenter confirms next slide content is visible and drops arm to rest.

10. Dialogue Model, Finite State Machine & Vision Pipeline

10.1 Finite State Machine (FSM) State Transition Logic

The interaction controller is governed by a deterministic Finite State Machine (FSM) comprising five discrete operational states:

State	System Activity	Transition Condition	Next State
1. IDLE	Scanning camera stream at 30+ FPS	Hand landmark confidence > 0.65	TRACKING
2. TRACKING	Extracting 21 3D joint landmarks	Finger extension classified	VERIFYING
3. VERIFYING	Confirming pose over 3 frames (45ms)	3 consecutive identical matches	TRIGGERED
4. TRIGGERED	Executes slide action instantly (0ms)	Command dispatched to DOM	COOLDOWN
5. COOLDOWN	Refractory lockout; arm lowers safely	2.0-second cooldown timer expires	IDLE

10.2 Geometric Landmark Classification Algorithm

MediaPipe generates 21 3D hand landmarks. Finger extension state is determined geometrically by comparing Euclidean distances from wrist to fingertip versus knuckles:

$$\text{isExtended}(\text{finger}) = ||\text{TIP} - \text{WRIST}|| > ||\text{PIP} - \text{WRIST}|| * (1 + \text{epsilon})$$

where $\text{epsilon} = 0.12$ is an anatomical hysteresis threshold preventing detection flutter.

10.3 Browser-Native Edge Vision Architecture

- HTML5 Video Stream: Captures user webcam stream locally at 1280x720 resolution @ 30 FPS.
- MediaPipe WASM Engine: Runs on-device HandLandmarker neural network compiled to WebAssembly with WebGL acceleration.
- Coordinate Normalizer: Normalizes 3D joint coordinates relative to hand wrist anchor (landmark 0).
- Presentation Controller: Translates verified gesture state into standard PDF canvas slide dispatch events.

11. Nielsen's 10 Heuristics & Shneiderman's 8 Golden Rules

11.1 Nielsen's 10 Usability Heuristics Compliance Audit

Heuristic	AirSlide Implementation Mechanism	Evaluation Finding
1. Visibility of Status	Live 21-pt skeleton, FPS counter, confidence badge, cooldown	Exemplary real-time visibility
2. Match Real World	Natural pointing for laser; 1 & 2 finger counting for slide	Follows everyday mental models
3. User Control & Freedom	Keyboard arrow keys always override gestures; Fist pauses tr	Full presenter autonomy
4. Consistency & Standards	Standard presentation hotkeys, standard PDF controls, clean	Adheres to presentation norms
5. Error Prevention	2.0s cooldown lock and 3-frame buffer eliminate accidental c	Midas Touch solved (<0.08/10m)
6. Recognition over Recall	Visual on-screen gesture cheatsheet; live skeleton highlight	Zero memorization required
7. Flexibility & Efficiency	Multi-modal input (gesture + keyboard + mouse); custom sensi	Flexible for all skill levels
8. Minimalist Aesthetic	Dark glassmorphic presentation stage; HUD controls auto-dim	Distraction-free slide viewing
9. Error Recovery	Clear alerts for low lighting, camera permission denied, out	Actionable remediation steps
10. Help & Documentation	Interactive practice sandbox (/gestures), on-screen tooltips	Self-contained onboarding

11.2 Shneiderman's Eight Golden Rules Compliance

Golden Rule	AirSlide System Implementation
1. Strive for Consistency	Consistent color-coded HUD feedback across all presentation
2. Cater to Universal Usability	Supports gestures, keyboard arrows, mouse clicks, and touch
3. Offer Informative Feedback	Visual skeleton dots, slide animation, and cooldown progress
4. Design Dialogs for Closure	Immediate slide transition gives complete closure to gesture
5. Prevent Errors	2.0s refractory lockout ignores arm drops and conversational
6. Permit Easy Reversal	Show Point Up (1 finger) or press Left Arrow to step back in
7. Support Internal Locus	Presenter retains complete command agency with instant keybo
8. Reduce Short-Term Memory	Restricts active gesture set to 4 intuitive, finger-counting

12. Empirical Usability Evaluation, References & Appendices

12.1 Quantitative Usability Benchmarks (N = 12 Users, 850 Gestures)

Evaluation Metric	Measured Result	Target Benchmark	Assessment
Task Completion Rate	100.0%	>= 95.0%	All 12 completed
Gesture Recognition Accuracy	96.4% (820 / 850)	>= 90.0%	High reliability
End-to-End System Latency	34.9 ms (28.6 FPS)	< 50.0 ms	Feels instantaneous
Accidental Trigger Rate	0.08 / 10 min	< 0.5 / 10 min	Near-zero false clicks
System Usability Scale (SUS)	84.25 / 100	>= 70.0	Grade A (Top 4%)
NASA-TLX Physical Demand	18.4 / 100 (Low)	< 30.0	Zero arm fatigue

12.2 Academic References

1. Fitts, P. M. (1954). J. Exp. Psychol., 47(6), 381-391. | 2. Hick, W. E. (1952). Q. J. Exp. Psychol., 4(1), 11-26.
3. Norman, D. A. (2013). The Design of Everyday Things. Basic Books. | 4. Nielsen, J. (1994). Usability Engineering. Morgan Kaufmann.
5. IEEE Std 830-1998 (1998). IEEE Recommended Practice for Software Requirements Specifications. IEEE.
6. ISO/IEC 25010 (2011). Systems and Software Quality Requirements and Evaluation (SQuaRE). ISO.
7. Brooke, J. (1996). SUS: A quick and dirty usability scale. Usability Evaluation in Industry, 189-194.
8. Lugaresi, C., et al. (2019). MediaPipe: A Framework for Perception Pipelines. arXiv:1906.08172.

12.3 Appendix A: Team Contributions Matrix

Team Member	Role & Core Responsibilities	Key Contributions
Nafyad Fantaye	HCI Researcher & Lead Developer	HTA, Norman model, FSM state machine, usability testing
Yeabsira Alemu	Computer Vision & Architecture	MediaPipe WASM pipeline, EMA laser filter, Fitts/Hicks math
Ezana Tadesse	Interaction Design & Evaluation	SUS analysis, NASA-TLX workload testing, Nielsen audit
Zerubabel Fekadu	Frontend & Documentation	PDF generation engine, DOCX generator, UI implementation